

INTRODUCTION

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Learning Objectives

1. Introduction
2. Computers and Learning
3. Types of Learning
4. Common Types of ML Problems
5. Learning Stages
6. ML Pipeline

Introduction

- Machine Learning(**ML**) is the science of programming computers so that they can learn from data.
 - Machine Learning(**ML**) is programming computers to optimize a performance criterion using example data or past experience.
 - **ML** concept gives computers the ability to think and helps us to solve many problems.
 - **ML** is about learning from existing data to make predictions about the future. This mainly deals with the extraction of patterns from a dataset.
 - For this, large datasets are used to identify (infer) patterns/insights and make decisions (predictions).
 - It's based on creating ML models from input data sets for data-driven decision making. These models are not just optimal but they can adapt to the changes in the environment.
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Systems & Learning

Learning is used when:

- Humans are unable to find patterns in huge datasets
 - Human expertise does not exist (navigating on Mars)
 - Humans are unable to explain their expertise (speech recognition)
 - Solution changes in time (routing a computer on a network)
 - Solution needs to be adapted to particular cases (user biometrics)
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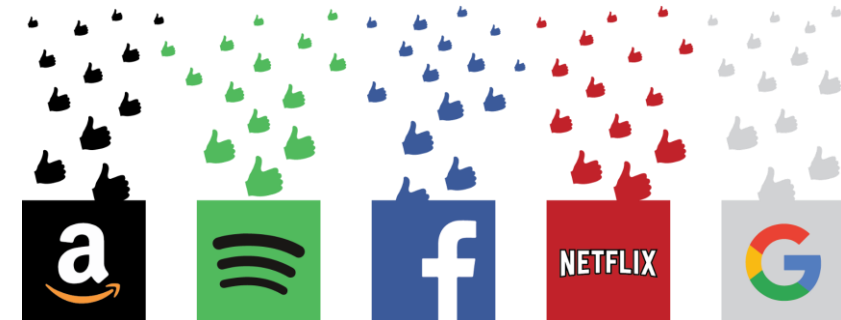
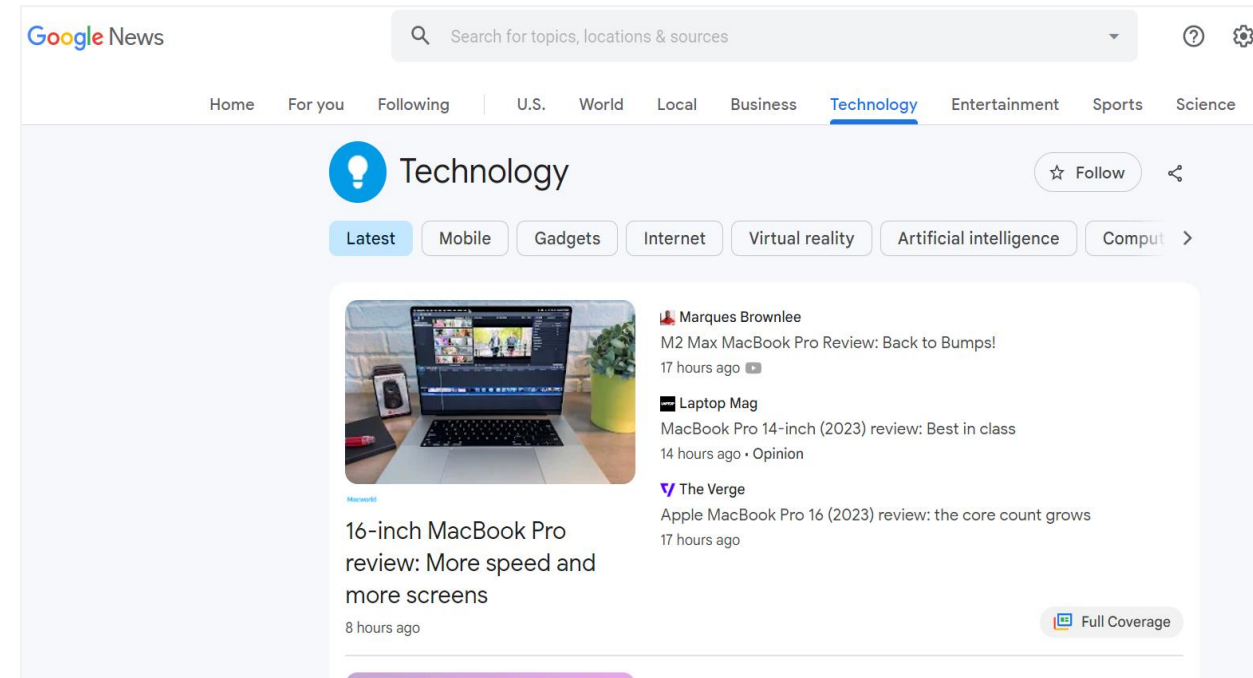
Machine Learning

With Machine Learning:

- We can **teach a system** from a dataset and let the system act by itself to **predict** future outcomes.
 - Write a program for language translation and use **ML model** to make the program work on other **different language data** to get the same results.
 - ML models can **scale up** to massive data volumes.
 - **Automated decision-making** is what makes ML so appealing.
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Examples- ML

- Predict the topic of an unseen document
- Classify the News article title
- Credit Card
- Weather & Stock predictions
- Recommender systems Ex. Netflix, E-commerce
- Email Spam/Not Spam.



Learning Associations

If there is a certain observed behavior, typically what is in association with that particular behavior

Analysis:

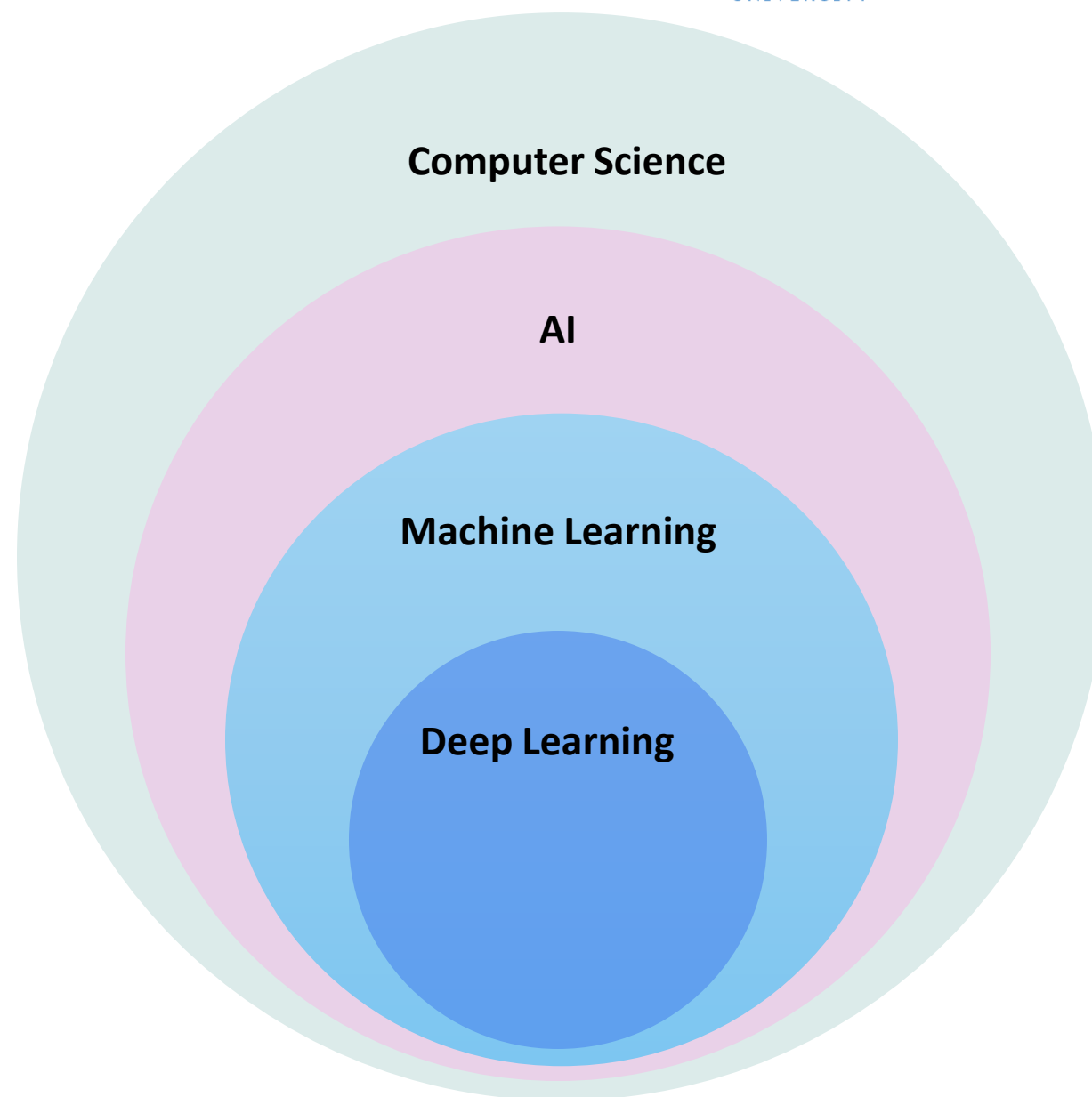
$P(Y|X)$ probability that a customer who buys X also buys Y where X and Y are products or services.

Example: $P(\text{Airpods} | \text{iphone}) = 0.7$

A customer who buys X but not Y is a potential customer for Y and can be targeted with advertisements.

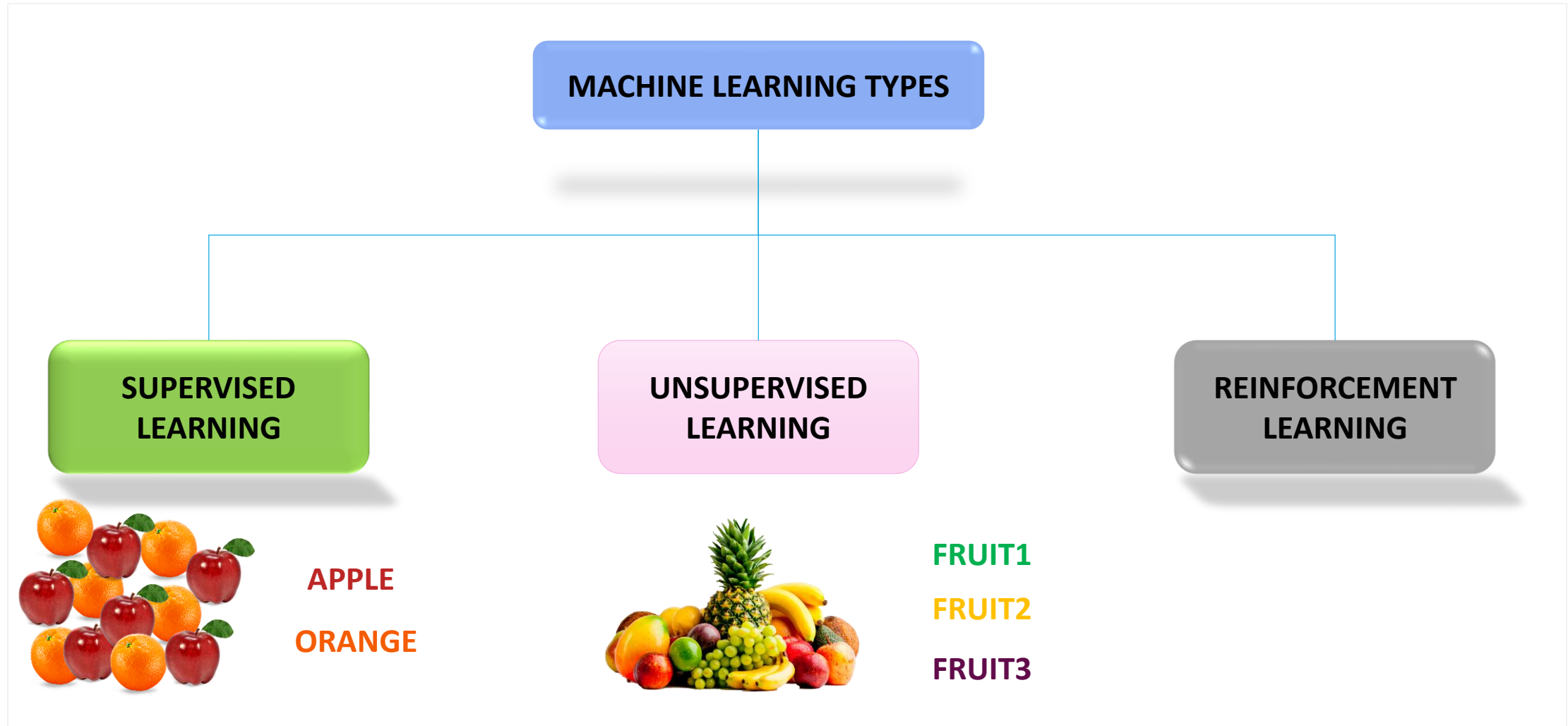
AI/ML/DL

- AI – Artificial intelligence leverages computers and machines to mimic the problem-solving and decision-making capabilities of the human mind
- ML – **ML** is a subset of artificial intelligence focused on using algorithms that learn and improve without being explicitly programmed to do so.
- DL – Deep Learning is a subset of machine learning, which is essentially a neural network with three or more layers. These neural networks attempt to simulate the behavior of the human brain—albeit far from matching its ability—allowing it to “learn” from large amounts of data.



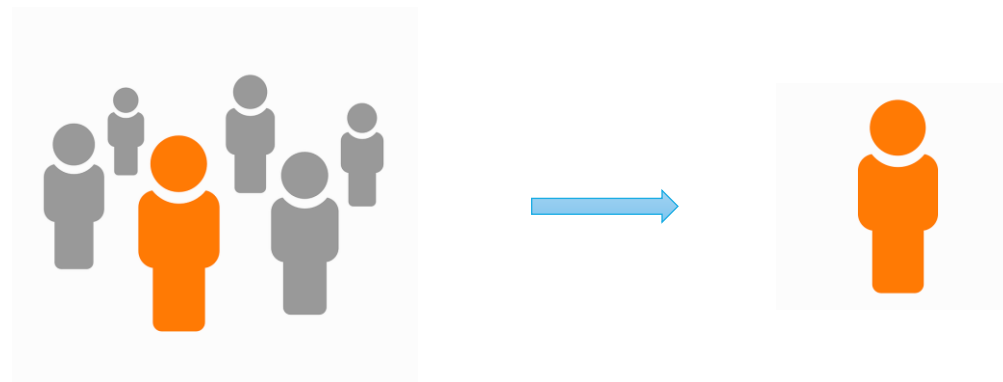
LEARNING TYPES

Learning Types



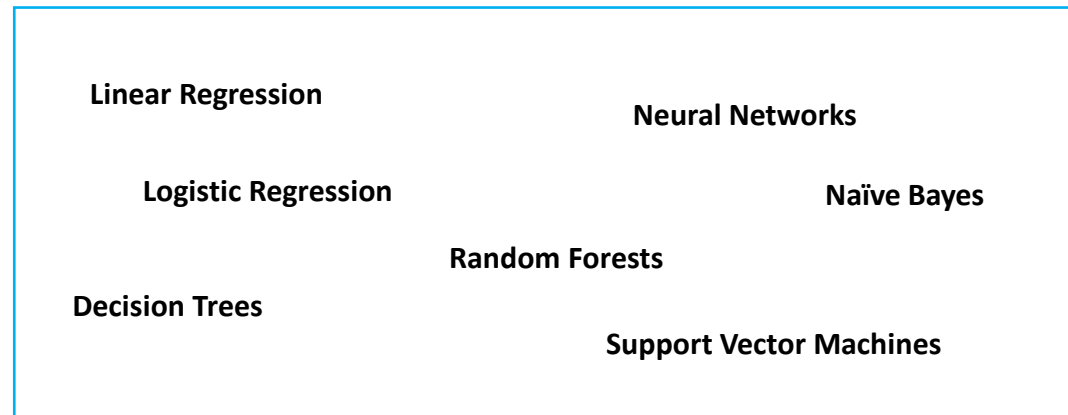
Supervised Learning

- The goal of a supervised learning algorithm is to use a dataset to build an ML model that takes a **feature vector x** as input.
- Then, it gives **labels as output** information deduced for this feature vector.
- For instance, the model created using the **dataset of people** could take a feature vector describing a **person as input** and output a probability that the person has an **illness**.



Supervised Learning

- Learning a **function** that maps from **x** to **y**, using labeled training examples **(x,y)**.
- Supervised learning algorithms include **linear regression**, **logistic regression**, Decision Trees, Neural networks etc.
- There are many forms of machine learning, but the majority of ML practical value today comes from **Supervised Learning**.



Example - Supervised Learning

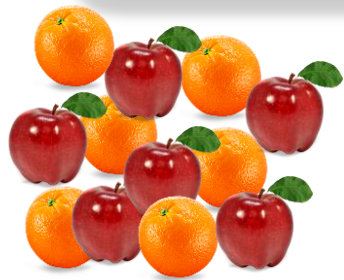
- We have a few emails which are marked Spam and Not Spam.
- This labeled data is used for supervised learning i.e., this data is used to train the model.
- After training the model, we can test it with new emails and verify if the model is able to correctly predict the new incoming email as Spam or Not Spam



Supervised Learning Types

SUPERVISED LEARNING

CLASSIFICATION



APPLE

ORANGE

REGRESSION

90% accurate					80% accurate		50% accurate		
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed
							?	?	?
76°	74°	70°	70°	71°	76°	75°			

Supervised Learning - **CLASSIFICATION**

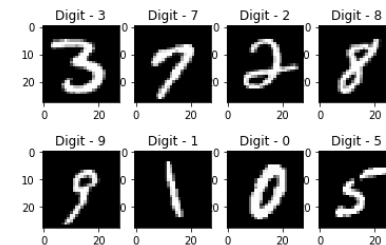
- If the output is categorical, the task is called classification.

Example:

1. Classification of cats and dogs.
2. Email – Spam/Not Spam

Applications

- Face Recognition: Emotion, Pose, Occlusion
- Handwriting digits (MNIST)
- Medical Diagnosis: From symptoms to illnesses
- Biometrics: Authentication/Recognition



Supervised Learning - **REGRESSION**

If the output is a continuous numerical value, then the ML approach is called Regression.

Example:

1. Predicting temperature (weather)
2. Stock Prices

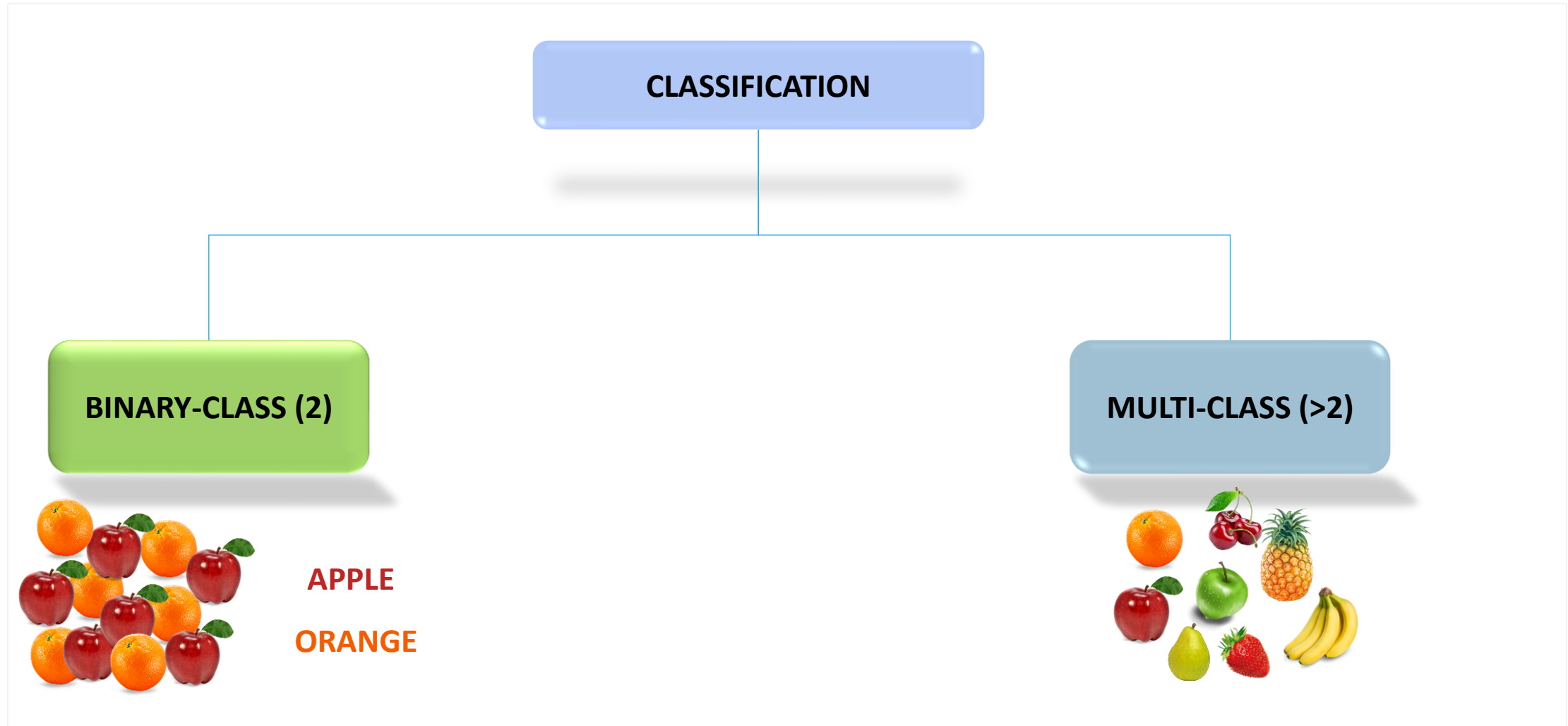
Applications

- Forecast Trends (Finance)
- Predict Outcomes (Prices/Weather)
- House Prices

90% accurate					80% accurate		50% accurate		
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed
							?	?	?
76°	74°	70°	70°	71°	76°	75°			



CLASSIFICATION - CLASSES



Binary Classification

Binary classification is the task of classifying the output of a prediction into two groups (each called class/category) on the basis of a classification rule.

The output of a class/category or label is based on the prediction result by a model.

These categories are predefined and is used to classify new probabilistic observations into any one of the categories.

0	1
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Yes	No
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Cats	Dogs
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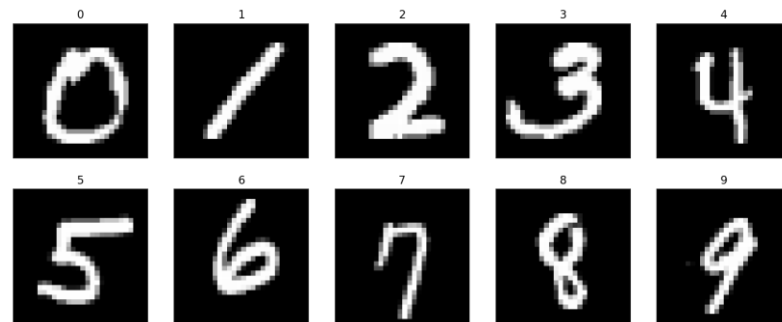
High	Low
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True	False
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Spam	Not Spam
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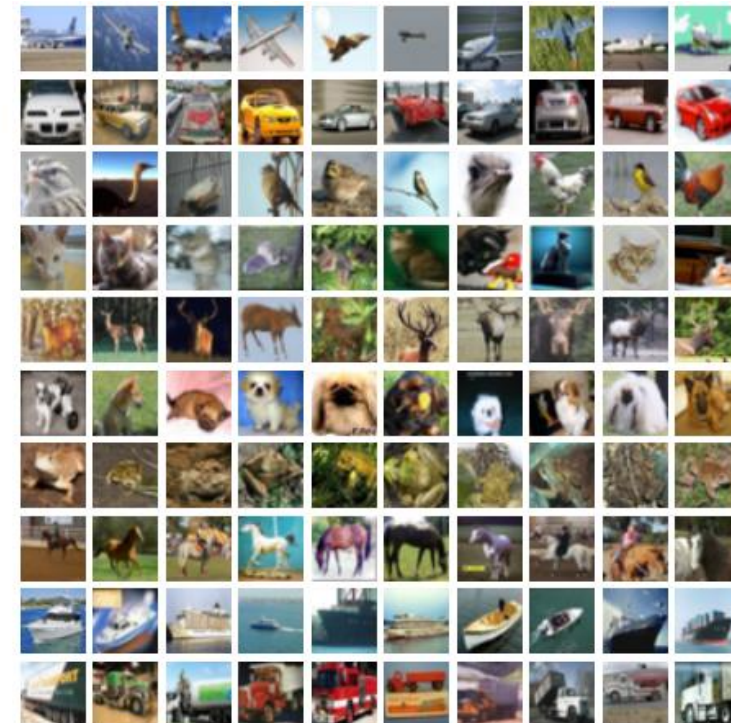
Multi-Class Classification

In machine learning, multiclass classification or multinomial classification is the problem of classifying instances into one of three or more classes.



MNIST Digits

airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck



CIFAR-10

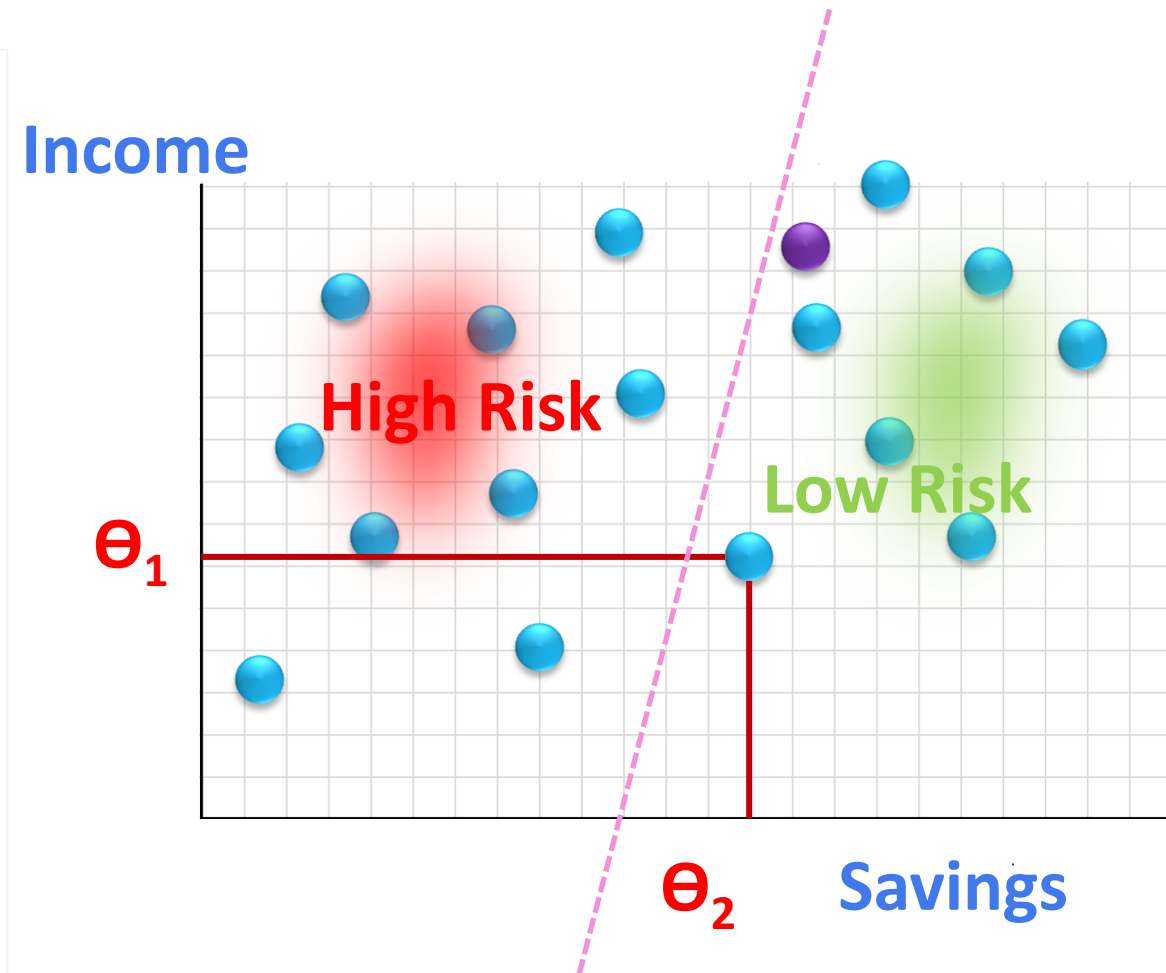
PROBLEM STATEMENTS IN SUPERVISED LEARNING

SL - Classification Task

Credit Scoring – Financial Services

- Differentiating between low-risk and high-risk customers from their income and savings
- Discriminant:
IF income $> \Theta_1$ AND savings $> \Theta_2$
THEN low-risk
ELSE high-risk

This belongs to the Classification problem type.



SL - Regression

Predicting the price of a used car

Different input features(car attributes) are mapped to the output price. Once the ML model is trained, then it can predict the prices based on the car attributes

X : Car attributes

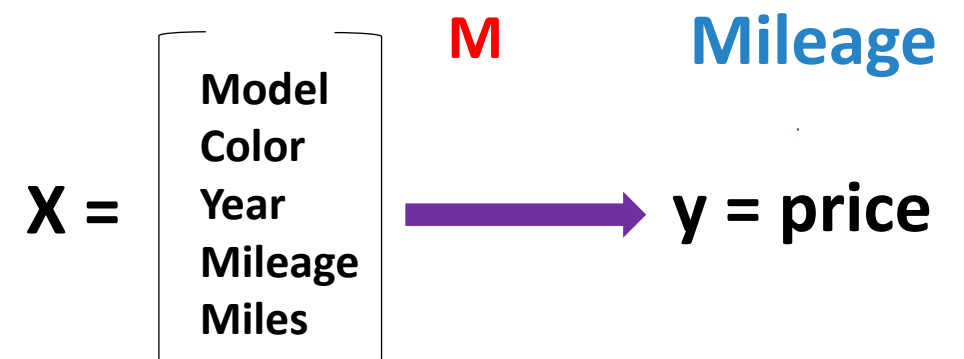
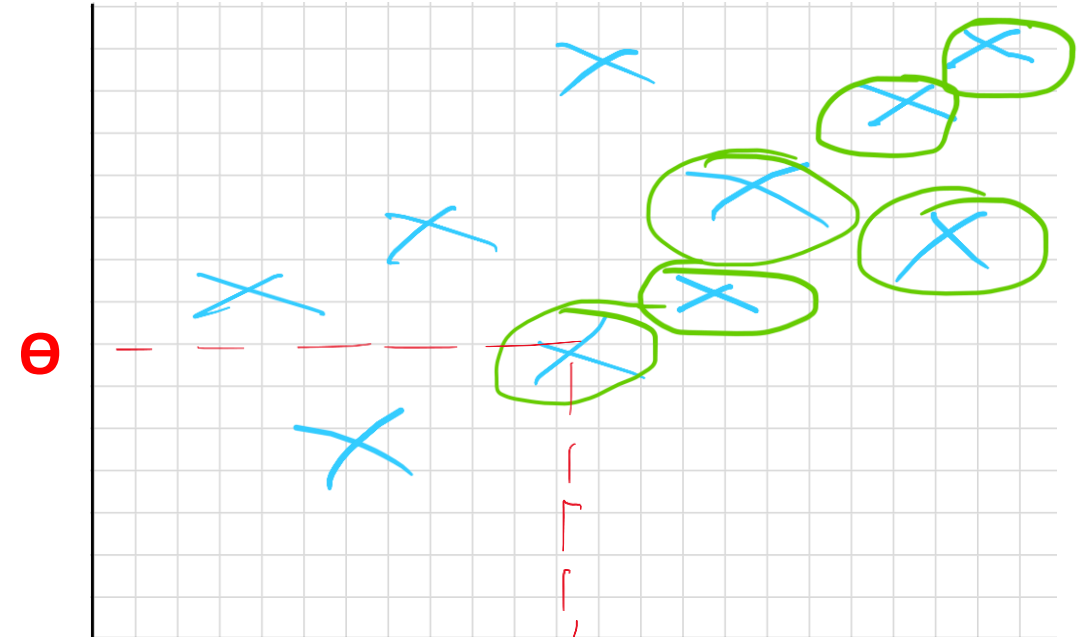
Y : Price

$$y = g(x | \Theta) \text{ or } y = g_{\Theta}(x)$$

$g()$, model (fixed)

Θ , parameters (modifiable $\rightarrow$ learned)

Price



Unsupervised Learning

Unsupervised Learning directly infers the properties without knowing the “correct” answers or the error for each observation.

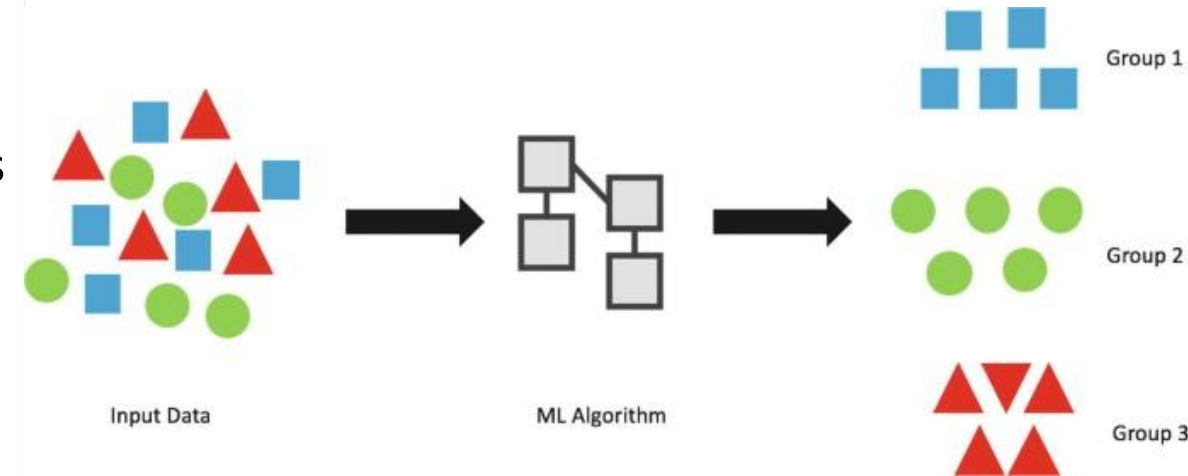
This is used to **find hidden patterns** and correlations within the raw data.

No training data is used in this model, so this technique is based on **unlabeled data**.

- The Learning is what normally happens
- No output
- Clustering: Grouping similar instances

Applications:

1. Customer Segmentation
2. Community Detection
3. Feature Selection
4. Bioinformatics: Learning Motifs

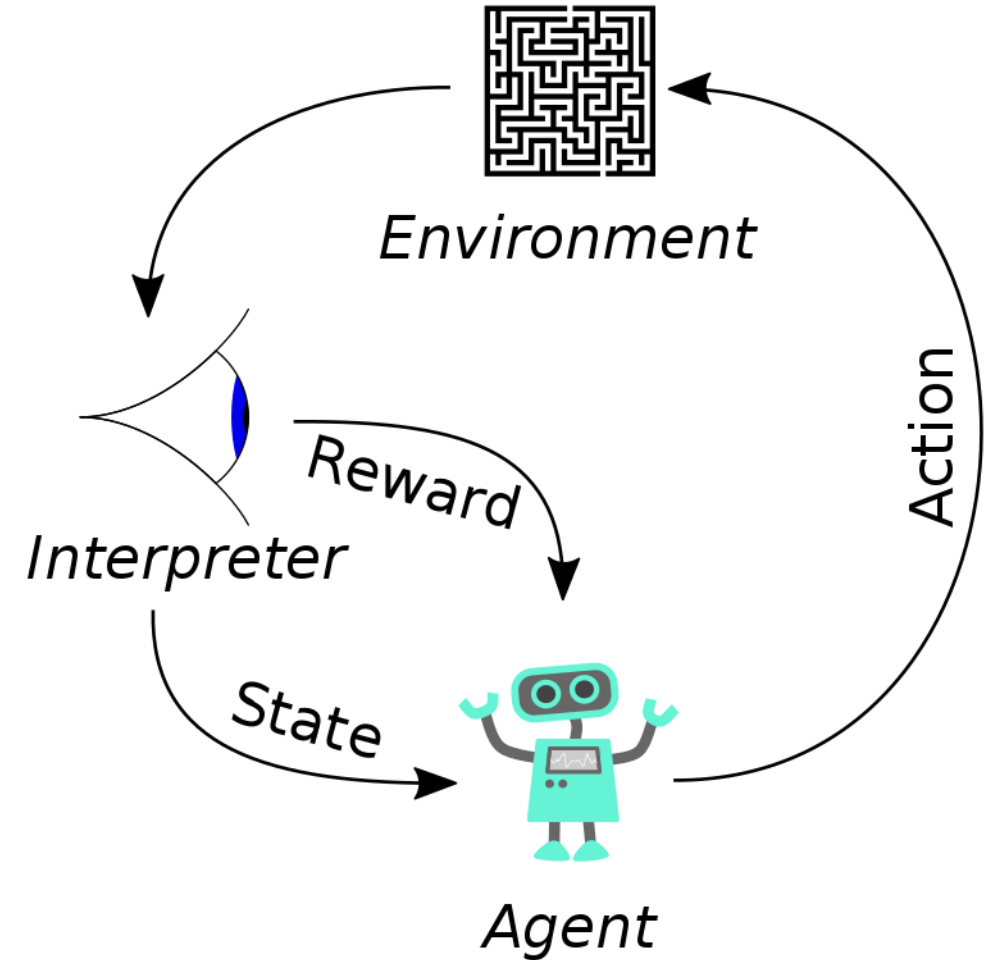


Reinforced Learning

- Learning a policy: A sequence of outputs
- No supervised output but delayed reward

Examples:

- Credit assignment problem
- Game Playing
- Robot in maze
- Multiple agents, partial observability



END
